Nguyen Tam Thang

AI Engineer

077-992-3158 | tamthang0512@gmail.com | linkedin.com/in/thangnguyen0512 | github.com/Thazg

Summary

Second-year Computer Science student at Ho Chi Minh City University of Technology (HCMUT), Vietnam National University Ho Chi Minh City, focused on AI engineering, recommender systems, and information retrieval. Hands-on experience building and evaluating recommendation and RAG pipelines, developing FastAPI services, and validating systems through reproducible benchmarks and automated tests.

Education

Ho Chi Minh University of Technology (HCMUT) — Ho Chi Minh City, Vietnam

Bachelor of Computer Science, GPA: 4.0/4.0 — *Sep. 2025 – Nov. 2029 (Expected)*

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Linear Algebra, Probability & Statistics, Computer Architecture

Projects

Enterprise Knowledge Intelligence Platform | *Python, FastAPI, Qdrant, Docker, RAG*

- Built an end-to-end RAG platform covering document ingestion, chunking, BGE embeddings, Qdrant indexing, hybrid Dense + BM25 retrieval, grounded generation, and citations
- Benchmarked Dense, BM25, Weighted RRF, Multi-Query, and reranking strategies on 100 queries; selected Weighted RRF with 0.7247 MRR and 0.9300 Hit@10
- Designed a lightweight cloud retrieval profile using FastEmbed and exact BM25-compatible sparse vectors, achieving 100/100 exact top-10 ranking parity with the local BM25 implementation
- Added API validation, dependency-aware readiness, generation timeouts, admission control, request IDs, Prometheus metrics, and automated CI/regression tests
- Deployed the system publicly with Render, Qdrant Cloud, and Groq using a 130.5 MB Docker image within a 512 MiB memory budget

E-commerce Recommendation System | *Python, implicit ALS, FastAPI, Docker*

- Built a nine-fold rolling benchmark on 2.76 million Retailrocket events with train-time catalogs, seen-item filtering, and separate warm/cold cohorts
- Combined ItemKNN, implicit ALS, and popularity sources through reciprocal rank fusion, then trained a learned ranker over the shared candidate pool
- Improved HR@20 from 3.8518% to 5.4566% (+41.7% relative) across 59,323 warm user-windows; folds 5–9 are post-selection diagnostics, not blind validation
- Packaged a versioned model bundle and local FastAPI service with input guards, caching, and metrics; preserved 101/101 reference outputs before and after restart

Technical Skills

Languages: Python, SQL, C/C++, Java
AI/ML & Retrieval: PyTorch, implicit ALS, ItemKNN, Learned Ranking, BGE Embeddings, BM25, Hybrid Search, RRF, SentenceTransformers, FastEmbed
Frameworks & Databases: FastAPI, Qdrant, Pydantic
Tools & Infrastructure: Git, Docker, Docker Compose, GitHub Actions, Prometheus, Grafana, Ollama, Render

Certifications

IELTS Academic: Overall Band 6.5 (CEFR B2)